Wall time per step: Splat simulation, 500k (482,409 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	44	28	25	23	OOM	11	21	205	23	225	41	66
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	14	13	4.0
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.9	0.8	1.9
hvg	0.7	0.8	0.7	0.6	–	3.1	4.5	0.5	0.5	19	19	13
pca	3.6	3.8	1.3	1.7	–	10	11	14	14	OOM	OOM	271
harmony	1.7	1.6	1.1	0.6	–	2.1	2.0	100	84	–	–	88
knn_snn	14	14	7.3	6.6	–	33	29	40	40	–	–	45
cluster	4.0	4.0	3.3	3.2	–	20	20	116	110	–	–	132
cluster_fast	14	15	3.1	2.5	n/a	n/a	n/a	196	194	n/a	n/a	n/a
umap	33	33	3.8	3.5	–	226	226	213	219	–	–	256
dge	1.8	1.9	1.7	1.7	–	454	–	22	22	–	–	404
tsne_default	247	245	217	211	–	1,315	1,319	n/a	n/a	–	–	1,404
tsne_fft	64	61	13	20	–	153	156	n/a	n/a	–	–	157
tsne_quick	41	41	25	23	n/a	n/a	n/a	155	152	n/a	n/a	n/a
total (no tSNE)	103	87	45	41	failed	766	317	712	515	failed	failed	1,281

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.